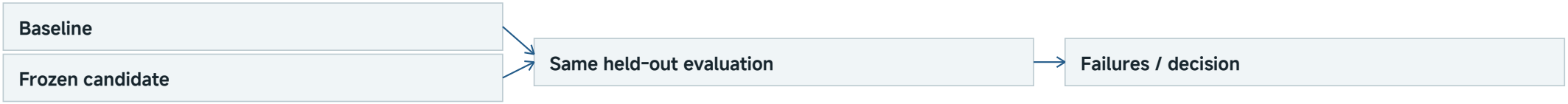
05 / Learning

How is experience used, and what contribution is supported?

EX06 / NeuralFeels | Source analysis | C1 | P: paper; W: project; R: repository



1. Role of experience

State how interaction data support training, inference, evaluation or design. No new training is required; explain your scope.

Pretrained touch-depth frontend differs from online object-field/pose fitting. An existing rotation policy produces observations; perception improvement is not a newly trained controller. [P Section 4; S2]

2. Data and independence

Identify data origin, selection and development/test groups. Keep related episodes together; disclose tuning or leakage.

Separate frontend training, per-object online fitting and evaluation sequences. Reusing a sequence over five seeds measures estimator variability, not new physical coverage. [P Fig 3]

3. Comparison and limits

Compare frozen methods or data strategies under matched conditions and budgets. Include denominators and all failure outcomes.

Same-sequence vision/fusion comparison: physical pose drift improves by 26.6%; simulated by 21.3%. Known-CAD tracking and heavy-occlusion ablations must retain their own scope. [P 2.2-2.4]

4. Supported contribution

Retain, revise or reject the bounded contribution. No improvement is valid; an unfinished evaluation is not a negative result.

Support bounded perception improvement. Analyst next checks one failure and its reference trace; better pose estimation alone does not prove higher manipulation-task success.

Review: Would the apparent improvement survive the same reset, action limits and trial budget for both methods?

Example response: Online field fitting, frontend pretraining and policy learning must not be merged into one training claim.

Worksheet 5 / Learning - evidence and team review

EX06 / NeuralFeels | SOURCE ANALYSIS EXAMPLE | Original review-page layout

Example ID: EX06 Project: NeuralFeels Claim ID: C1 Revision: draft 2

Mode: Source analysis Pages: canvas + this review; Decision also includes both A3 sides

Evidence scope: selected paper text/tables checked 20 Oct 2026; no raw dataset download or independent experiment.

Evidence ID	Artifact/source + location (clickable)	What it supports; limitation / access gap
P	arxiv.org/html/2312.13469v1	Paper: method/results; reported evidence, not our measurement.
W	suddhu.github.io/neural-feels	Project: system/videos; selected examples do not establish reliability.
R	github.com/facebookresearch/neuralfeels	Repository: linked route; code, raw data and current commit not audited.

Next action / owner / verification:

Analyst: separate pretraining, online fitting and estimator seeds; inspect one failed or heavily occluded case.

Teaching example: no actual student team or signatures. Source analysis is not a reproduced result; blank signature fields are intentional.

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P/W/R identify linked public sources above. NE: not examined in this brief example, not evidence of failure. Support/context labels are provisional, not grades.